**Q1 - Single Correct**

If $m/n$ (in lowest term) be the probability that a randomly chosen positive divisor of $10^{99}$ is an integral multiple of $10^{88}$, then the value of $(m+n)$ is

(1) 634

(2) 638

(3) 616

(4) None of these

**Q2 - Single Correct**

A determinant is chosen at random from the set of all determinants of order 2 with elements 0 or 1 only. The probability that the determinant chosen has the value non-negative, is

(1) $3/16$

(2) $6/16$

(3) $10/16$

(4) $13/16$

**Q3 - Single Correct**

An urn contains 10 balls coloured either black or red. When selecting two balls from the urn at random, the probability that a ball of each colour is selected is $8/15$. Assuming that the urn contains more black balls than red balls, then the probability that atleast one black ball is selected, when selecting two balls, is

(1) $18/45$

(2) $30/45$

(3) $39/45$

(4) $41/45$

**Q4 - Single Correct**

A fair die is tossed repeatedly. Mr. A wins if it is 1 or 2 on two consecutive tosses and Mr. B wins, if it is 3, 4, 5 or 6 on two consecutive tosses. The probability that A wins, if the die is tossed indefinitely, is

(1) $1/3$

(2) $5/21$

(3) $1/4$

(4) $2/5$

**Q5 - Single Correct**

Mr. A makes a bet with Mr. B that in a single throw with two dice he will throw a total of seven before B throws four. Each of them has a pair of dice and they throw simultaneously until one of them wins, equal throws being disregarded. Probability that $B$ wins, is

(1) $1/3$

(2) $4/11$

(3) $5/16$

(4) $5/17$

**Q6 - Single Correct**

If $a, b$ and $c$ are three numbers (not necessarily different) chosen randomly and with replacement from the set $\{1,2,3,4,5\}$, the probability that $(ab+c)$ is even, is

(1) $35/125$

(2) $59/125$

(3) $64/125$

(4) $75/125$

**Q7 - Single Correct**

Mr. A and Mr. B each have a bag that contains one ball of each of the colours blue, green, orange, red and voilet. 'A' randomly select one ball from his bag and puis it into B's bag. B' randomly select one ball from his bag and puts it into A's bag. The probability that after this process the contents of the two bags are the same, is

(1) $1/6$

(2) $1/5$

(3) $1/3$

(4) $1/2$

**Q8 - Single Correct**

Two boys $A$ and $B$ find the jumble of $n$ ropes lying on the floor. Each takes hold of one loose and randomly. If the probability that they are both holding the same rope is $\frac{1}{101}$ Then, the number of ropes is equal to

(1) 101

(2) 100

(3) 51

(4) 50

**Q9 - Single Correct**

A letter is taken at random from the letters of the word 'STATISTICS' and another letter is taken at random from the letters of the word 'ASSISTANT'. The probability that they are same, is

(1) $1/45$

(2) $13/90$

(3) $19/90$

(4) $5/18$

**Q10 - Single Correct**

A fair coin is tossed 7 times, then probability that atleast 3 consecutive heads occurs is equal to

(1) $1/8$

(2) $3/8$

(3) $5/8$

(4) $3/7$

**Q11 - Single Correct**

Let $S=\{a \in N, a \leq 1000\}$. If equation $\left[\tan^2 2x\right]-\tan 2x-a=0$ has real roots, then probability of getting element 'a' from the set(s), which satisfy the equation

(1) $p=25/999$

(2) $p=125/999$

(3) $p=31/1000$

(4) $p=27/1000$

**Q12 - Single Correct**

Three numbers are drawn from the set $\{1,2,3,\ldots,n\}$ with replacement. The probability that their sum is $2n$, is

(1) $1/2n$

(2) $1/2$

(3) $\frac{(n-1)(n+4)}{2n^3}$

(4) $\frac{(n+1)(n+2)}{n^3}$

**Q13 - Multiple Correct**

Two whole numbers are randomly selected and multiplied. Consider two events $E_1$ and $E_2$ defined as

$E_1$ : Their product is divisible by 5.

$E_2$ : Units place in their product is 5.

Which of the following statements(s) is/are correct?

(1) $E_1$ is twice as likely to occur as $E_2$

(2) $E_1$ and $E_2$ are disjoint

(3) $P\left(E_2/E_1\right)=1/4$

(4) $P\left(E_1/E_2\right)=1$

**Q14 - Multiple Correct**

If $A$ and $B$ are two events such that $P(B) \neq 1, B^C$ denotes the event complementary to $B$, then

(1) $P\left(A / B^C\right)=\frac{P(A)-P(A \cap B)}{1-P(B)}$

(2) $P(A \cap B) \geq P(A)+P(B)-1$

(3) $P(A)><P(A / B)$ according as $P\left(A / B^C\right)><P(A)$

(4) $P\left(A / B^C\right)+P\left(A^C / B^C\right)=1$

**Q15 - Multiple Correct**

Which of the following statements(s) is/are correct?

(1) 3 coins are tossed once. Two of them atleast must land the same way. No matter whether they land heads or tails, the third coin is equally likely to land either the same way or oppositely. So, the chance that all the three coins land the same way is $1/2$.

(2) Let $0<P(B)<1$ and $P(A / B)=P\left(A / B^C\right)$, then $A$ and $B$ are independent.

(3) Suppose an urn contains 'w' white and ' $b$ ' black balls and a ball is drawn from it and is replaced along with 'd' additional balls of the same colour. Now, a second ball is drawn from it The probability that the second drawn ball is white is independent of the value of 'd'

(4) $A, B, C$ simultaneously satisfy $P(ABC)=P(A) \cdot P(B) \cdot P(C)$ and $P(AB\bar{C})=P(A) \cdot P(B) \cdot P(\bar{C})$ and $P(A\bar{B}C)=P(A) \cdot P(\bar{B}) \cdot P(C)$ and $P(\bar{A}BC)=P(\bar{A}) \cdot P(B) \cdot P(C)$. Then, $A, B, C$ are independent.

**Q16 - Multiple Correct**

Sixteen players $S_1, S_2, \ldots, S_{16}$ play in a tournament. They are divided into eight pairs at random. From each pair, a winner is decided on the basis of a game played between the two players of the pair. Assume that all the players are of equal strength. The probability that 'exactly one of the two players $S_1$ and $S_2$ is among the eight winners', is

(1) $4/15$

(2) $7/15$

(3) $8/15$

(4) $9/15$

**Q17 - Multiple Correct**

The number a is randomly selected from the set $\{0, 1, 2, 3, \ldots, 98, 99\}$. The number $b$ is selected from the same set. Probability that the number $3^a + 7^b$ has a digit equal to 8 at the unit place, is

(1) $1/16$

(2) $2/16$

(3) $4/16$

(4) $3/16$

**Q18 - Multiple Correct**

A signal which can be green or red with probability $4/5$ and $1/5$ respectively, is received by station $A$ and then transmitted to station $B$. The probability of each station receiving the signal correctly is $(3/4)$. If the signal received at station $B$ is green, then the probability that the original signal was green, is

(1) $3/5$

(2) $6/7$

(3) $20/23$

(4) $9/23$

**Q19 - Multiple Correct**

A boy has a collection of blue and green marbles. The number of blue marbles to the sets $\{2, 3, 4, \ldots, 13\}$. If two marbles are chosen simultaneously and at random from his collection, then the probability that they have different colours is $1/2$. Possible number of blue marbles, is

(1) 2

(2) 3

(3) 6

(4) 10

**Q20 - Multiple Correct**

$A$ and $B$ are two inaccurate arithmeticians whose chance of solving a given question correctly are $1/8$ and $1/12$, respectively. They solve a problem and obtained the same result. If it is 1000 to 1 against their making the same mistake, then chance that the result is correct, is

(1) $\frac{13}{14}$
(2) $\frac{1}{14}$
(3) $\frac{1}{7}$
(4) $\frac{6}{7}$

**Q21 - Multiple Correct**

In a certain factory, machines $A, B$ and $C$ produce bolts of their production, machines $A, B$ and $C$ produce $2\%, 1\%$ and $3\%$ defective bolts, respectively. Machine A produces $35\%$ of the total output of bolts, machine $B$ produces $25\%$ and machine $C$ produces $40\%$. A bolt is chosen at random from the factory's production and is found to be defective. The probability it was produced on machine $C$, is

(1) $\frac{6}{11}$
(2) $\frac{23}{45}$
(3) $\frac{24}{43}$
(4) $\frac{3}{11}$

**Q22 - Multiple Correct**

Let $X$ and $Y$ be two events such that $P(X/Y) = 1/2, P(Y/X) = 1/3$ and $P(X \cap Y) = 1/6$. Which of the following is/are correct?

(1) $P(X \cup Y) = 2/3$
(2) $X$ and $Y$ are independent
(3) $X$ and $Y$ are not independent
(4) $P(\bar{X} \cap Y) = 1/3$

**Q23 - Multiple Correct**

Let $E$ and $F$ be two independent events. The probability that exactly one of them occurs is $11/25$ and the probability of none of them occurring is $2/25$. If $P(T)$ denotes the probability of occurrence of the events $T$, then

(1) $P(E)=4/5, P(F)=3/5$

(2) $P(E)=1/5, P(F)=2/5$

(3) $P(E)=2/5, P(F)=1/5$

(4) $P(E)=3/5, P(F)=4/5$

**Q24 - Paragraph 1**

**Passage I (For Question 24, 25)**

A box contains $b$ red balls, $2b$ white balls and $3b$ blue balls, where $b$ is a positive integer. 3 balls are selected at random from the box.

If balls are drawn without replacement and $A$ denotes the event that 'No two of the selected balls have the same colour', then

(1) there is no value of $b$ for which $P(A)=0.3$

(2) there is exactly one value of $b$ for which $P(A)=0.3$ and this value is less than 10

(3) there is exactly one value of $b$ for which $P(A)=0.3$ and they value is greater than 10

(4) there is more than one value of $b$ for which $P(A)=0.3$

**Q25 - Paragraph 1**

If balls are drawn without replacement and $B$ denotes the event that 'No two of the 3 drawn balls are blue', then

(1) $P(B)=1/3$, if $b=1$

(2) $P(B)=2/3$, if $b=2$

(3) $P(B)=1/4$, if $b=4$

(4) $P(B)=1/2$ for all value of $b$

**Q26 - Paragraph 2**

## Passage II (For Question 26, 27)

There are four boxes $A_1, A_2, A_3$ and $A_4$. Box $A_i$ has $i$ balls and on each ball a number is printed, the number are from 1 to $i$. A box is selected randomly, the probability of selection of box $A_i$ is $\left(\frac{i}{10}\right)$ and then a ball is drawn. Let $E_i$ represents the event that a ball with number ' $i$ ' is drawn.

The value of $\sum_{i=1}^{4} P\left(E_i \cap A_i\right)$ is

(1) $1/5$

(2) $1/10$

(3) $2/5$

(4) $1/4$

**Q27 - Paragraph 2**

$P\left(A_3 / E_2\right)$ is equal to

(1) $1/4$

(2) $1/3$

(3) $2/3$

(4) $1/2$

**Q28 - Paragraph 3**

## Passage III (For Question 28, 29)

$A$ and $B$ play a game in which they alternately call out positive integer less than or equal to $n$, according to the following rules:

A goes first and always calls out an odd number, $B$ always out an even number and each player must call out a number, which is greater than the previous number (except for A's first turn). The game ends when one player cannot call out a number.

For $n = 4$, number of games possible, is

(1) 3

(2) 4

(3) 6

(4) None of these

**Q29 - Paragraph 3**

For $n = 8$, number of games possible, is

(1) 20

(2) 21

(3) 4

(4) None of these

**Q30 - Paragraph 4**

## Passage IV (For Question 30, 31)

An urn contains 4 white and 9 black balls. $r$ balls are drawn with replacement. Let $P(r)$ be the probability that no two white balls appear in succession.

The value of $P(4)$ must be

(1) $\frac{81}{(169)^2}$

(2) $\frac{81 \times 273}{(169)^2}$

(3) $1 - \left(\frac{4}{13}\right)^4$

(4) None of these

**Q31 - Paragraph 4**

The recursion relation of $P_{(r)}$ must be

(1) $P_{(r)}=P_{(r-2)}\cdot\left(\frac{9}{13}\right)\left(\frac{4}{13}\right)+P_{(r-1)}\left(\frac{9}{13}\right)$

(2) $P_{(r)}=P_{(r-2)}\cdot\left(\frac{9}{13}\right)^2+P_{(r+1)}\left(\frac{9}{13}\right)$

(3) $P_{(r)}=P_{(r-2)}\cdot\left(\frac{4}{13}\right)^2+P_{(r-1)}\cdot\left(\frac{9}{13}\right)$

(4) None of the above

**Q32 - Paragraph 5**

## Passage V (For Question 32, 33)

Let $S$ be the set of students $\{a_1, a_2, \ldots, a_n\}$. The students plan to visit two institutes next month. Any student can accompany or with draw from the visit. So, on both occassions any subset of $S$ can go.

Probability that there is no common student in two visits, is

(1) $1/4^n$

(2) $(3/4)^n$

(3) $1/2^n$

(4) $3/4^n$

**Q33 - Paragraph 5**

Probability that each student either goes in first visit or in second or in both, is

(1) $1/4^n$

(2) $1/2^n$

(3) $(3/4)^n$

(4) $1$

**Q34 - Paragraph 6**

## Passage VI (For Question 34, 35)

A box contains nine slipe bearing the numbers $-4,-3-2,-1,0,1,2,34$ (one number for one slip).

A slip is chosen at random, its number is noted and replaced in the box. If the experiment is repeated 5 times, then the probability that exactly 3 numbers appear in the noted list, is

(1) $\frac{{}^9C_3 \times 90}{9^5}$

(2) $\frac{{}^9C_3 \times 243}{9^5}$

(3) $\frac{{}^9C_3 \times 150}{9^5}$

(4) $\frac{{}^9C_3 \times 240}{9^5}$

**Q35 - Paragraph 6**

If a slip is chosen at random from the box and the experiment is repeated nine times (with replacement) to form $3 \times 3$ matrix. The chosen number are arranged rowwise, then the probability that the matrix, so formed is skew-symmetric, is

(1) $(1/9)^8$

(2) $(1/9)^6$

(3) $(1/9)^3$

(4) $(1/9)^4$

**Q36 - Integer Type**

A positive integer that reads the same backwards and forwards is called a palindrome. A number greater than 10 and less than $10^6$ is chosen at random. If it is known that it is a five-digit number and the probability that it is a palindrome is $p$, then the value of $300p$, is

**Q37 - Integer Type**

A drawer contains several pair of socks. Not wanting to count Mr. A asked miss $X$ 'how many pairs of socks are there in the drawer'? Miss $X$, not wanting to give answer replied, 'well, each sock has exactly one matching sock and the probability that two socks drawn from the drawer from a matching pair is $(1/15)'$. Then, the answer to Mr. A's question, is

**Q38 - Integer Type**

If the papers of 4 students can be checked by one of the 7 teachers. If the probability that all the 4 papers are checked by exactly 2 teachers is $A$, then the value of 49 A must be

**Q39 - Integer Type**

A function is selected from all function of set $S=\{1,2,3,\ldots,n\}$ to itself. If the probability that it is one-one is $3/32$, then $n$ is equal to

**Q40 - Integer Type**

A bag contains $n$ identical red balls, $2n$ identical black balls and $3n$ identical white balls. If probability of drawing $n$ balls of same colour is greater than or equal to $1/6$, then the minimum number of red balls in the bag is equal to

**Q41 - Integer Type**

A coin that comes up with probability $p>0$ and tails with probability $1-p>0$ independently meach flip is flipped eight times. Suppose the probability of three heads and five tails is equal to $1/25$ of the probability of five heads and three tails. Let $p=\frac{m}{n}$, where $m$ and $n$ are relatively prime positive integers. The value of $(m+n-5)$ equals

**Q42 - Integer Type**

A man has 3 coins $A, B$ and $C$ out of which $A$ is fair. The probability of head appearing when tossed are $2/3$ and $1/3$ for $B$ and $C$, respectively. One of the coins chosen at random, is tossed three times giving 2 heads and one tail. If the probability that the chosen coin was $A$ is $k/25$, then $k$ is

**Q43 - Integer Type**

In a factory, which manufactures bolts, machines $A, B$ and $C$ manufacture respectively $25\%, 35\%$ and $40\%$ of the bolts of their outputs $5\%, 4\%$ and $2\%$ are respectively defective bolts. A bolt is drawn at random from the product and is found to be defective. The probability that it is manufactured by the machine $B$ is $\frac{m}{n}$. Then, $\frac{|2m-n|}{13}$ is equal to

**Q44 - Integer Type**

A bag $A$ contains 2 white balls, 3 red balls and a bag $B$ contains 4 white balls, 5 red balls. One ball is drawn at random from one of the bags and it is found to be red. The probability that it was drawn from bag $B$, is $\frac{m}{n}$. Then, $|2m-n|$ is equal to

**Q45 - Integer Type**

In a test, an examinee either guesses or copies or knows the answer to a multiple choice question with four choices. The probability that he makes a guess is $1/3$ and the probability that he copies the answer is $1/6$. The probability that his answer is correct given that he copied it, is $1/8$. The probability that he knew the answer to the question given that he correctly answered it is $m/n$. Then, $|m-n|$ is equal to

**Q46 - Integer Type**

A fair die is thrown 3 times. The chance that the sum of three numbers appearing on the die is less than 11, is equal to $\frac{m}{n}$, then $(m+n)$ is

# Answer Key

| | | | |
|---|---|---|---|
| **Q1** (1) | **Q2** (4) | **Q3** (3) | **Q4** (2) |
| **Q5** (1) | **Q6** (2) | **Q7** (3) | **Q8** (3) |
| **Q9** (3) | **Q10** (2) | **Q11** (3) | **Q12** (3) |
| **Q13** (3, 4) | **Q14** (1, 2, 3, 4) | **Q15** (2, 3, 4) | **Q16** (3) |
| **Q17** (4) | **Q18** (3) | **Q19** (2, 3, 4) | **Q20** (1) |
| **Q21** (3) | **Q22** (1, 2) | **Q23** (1, 4) | **Q24** (2) |
| **Q25** (4) | **Q26** (3) | **Q27** (2) | **Q28** (1) |
| **Q29** (2) | **Q30** (2) | **Q31** (2) | **Q32** (2) |
| **Q33** (3) | **Q34** (3) | **Q35** (2) | **Q36** (3) |
| **Q37** (8) | **Q38** (6) | **Q39** (4) | **Q40** (5) |
| **Q41** (6) | **Q42** (9) | **Q43** (1) | **Q44** (2) |
| **Q45** (5) | **Q46** (3) | | |